

Pressure and Fluid Statics

Part 2

INSTRUMENTS & MEASUREMENTS

Topic to cover

- Physical characteristics of fluid
- Mass, weight, density, specific gravity
- Pressure
- Hydrostatic pressure
- Pascal's Principle
- Gauge pressure, absolute pressure
- Pressure measurement
- Archimedes' Principle



Part 2

Absolute and Gauge Pressure

Absolute and gauge pressure

A pressure gauge is used to measure pressure. When a pressure gauge is not connected to anything, it gives a read of zero. But is it true that the pressure gauge is experiencing no pressure at all? What about the atmospheric pressure?



Source: Amazon.com

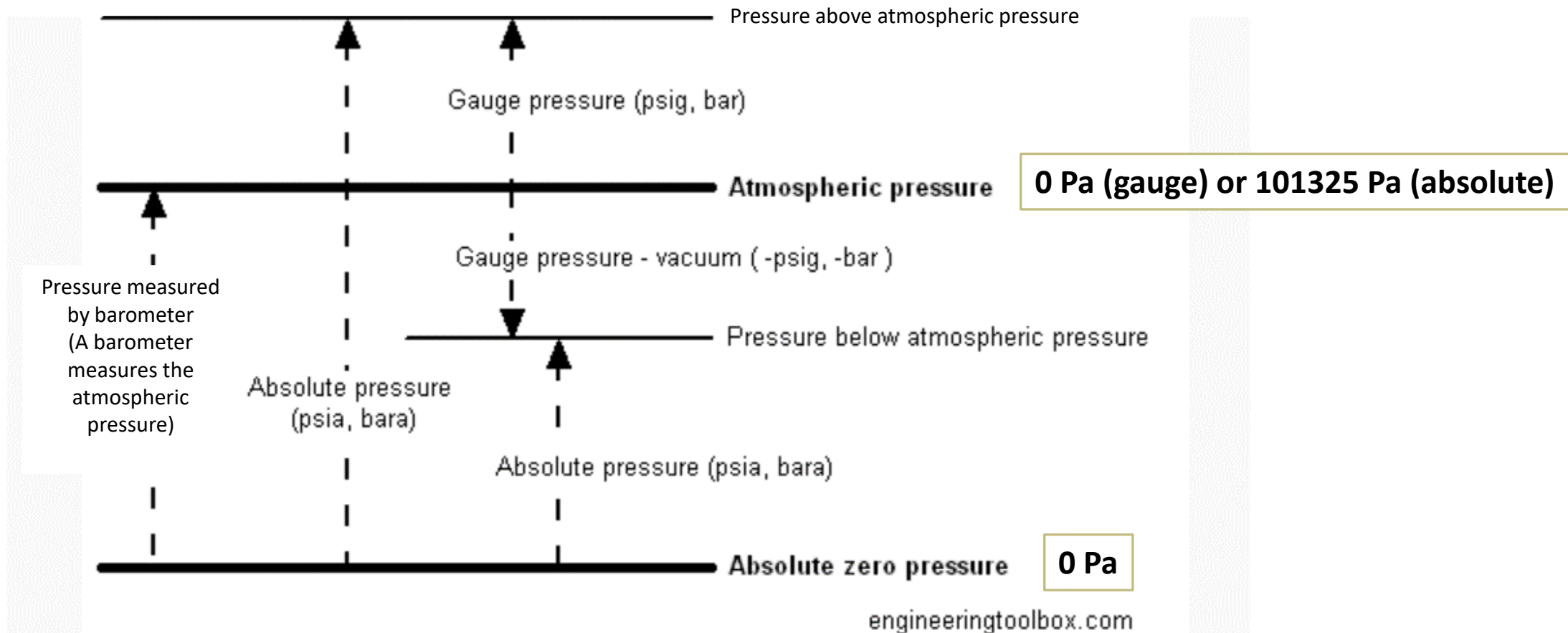
Absolute and gauge pressure

- Absolute pressure:
 - The measure of pressure with respect to absolute zero pressure, which is the pressure of a perfect vacuum*
 - By definition the smallest possible pressure is 0 Pa (abs)
- Gauge pressure:
 - The pressure relative to atmospheric pressure
 - Calculated by subtracting atmospheric pressure from absolute pressure.
 - Pressure gauge are designed to give a reading of zero gauge pressure when measuring atmospheric pressure

*Perfect vacuum is an ideal state with no particles at all

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmosphere}}$$

Absolute and gauge pressure

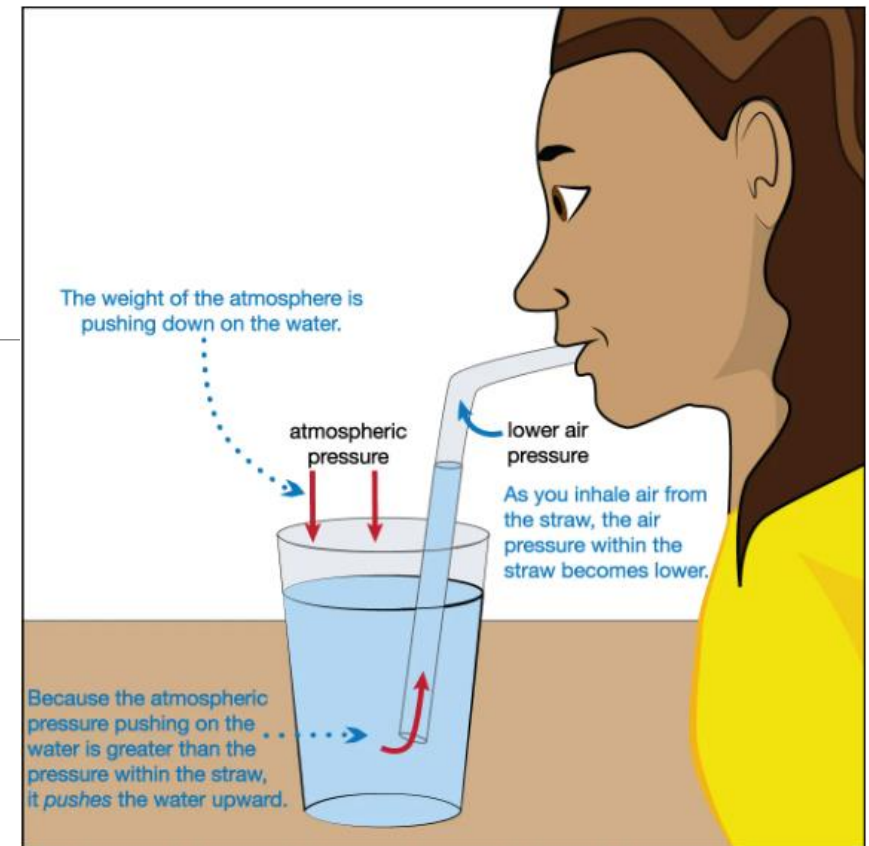


Vacuum

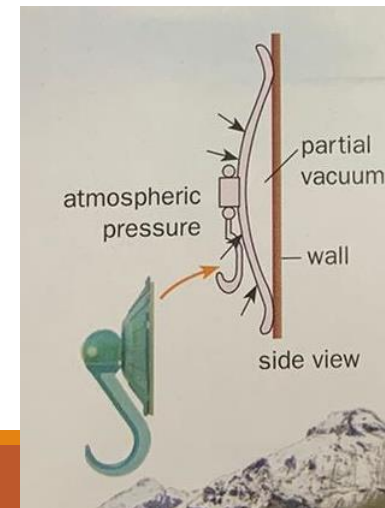
- A space that has a pressure $< P_{\text{atm}}$ is referred to as a vacuum or negative (gauge) pressure.
- When all the air is extracted, it creates a perfect vacuum ($P_{\text{abs}} = 0$) and is called ***absolute zero pressure***.
- This condition is never reached in practice although a pressure very near to the absolute zero pressure can be achieved.

Atmospheric pressure

- Pressure exerted by the weight of the atmosphere
- Specified as an absolute pressure (e.g. at sea level, $P_{\text{atm}} \approx 101325 \text{ Pa}$)
- Varies depending on altitude
- Air molecules around us take up space and have weight. Therefore the atmosphere exerts a pressure on the Earth's surface and on us. This pressure is known as atmospheric pressure.
- We are not crushed by atmospheric pressure because the natural pressure in our body is about the same as atmospheric pressure.
- Applications of atmospheric pressure:
 - Drinking from straw
 - Suction cups



Source: grade8science.com



Source: Physics Matters, 4th Edition

Example

Given that $P_{\text{atm}} = 101.3 \text{ kPa}$

1. If the measured pressure is 200.0 kPa (abs), what is the gauge pressure?
2. If the measured pressure is 200.0 kPa (g), what is the absolute pressure?
3. If the measured pressure is 101.3 kPa (abs), what is the gauge pressure?

Solutions

Given $P_{\text{atm}} = 101.3 \text{ kPa}$

1. If pressure is 200.0 kPa (abs), what is the gauge pressure?

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmosphere}} = 200.0 - 101.3 = 98.7 \text{ kPa(g)}$$

2. If pressure is 200.0 kPa (g), what is the absolute pressure?

$$P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atmosphere}} = 200.0 + 101.3 = 301.3 \text{ kPa(abs)}$$

3. If pressure is 101.3 kPa (abs), what is the gauge pressure?

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmosphere}} = 101.3 - 101.3 = 0.0 \text{ kPa(g)}$$

Example

1



Air pressure in basketball = 150 kPa (abs)

Atmospheric pressure = 101.3 kPa

Air pressure in basketball expressed in gauge pressure
 $= P_{\text{abs}} - P_{\text{atm}} = 150 - 101.3 = 48.7 \text{ kPa(g)}$

Basketball (located at sea level)

2



Air pressure in packaging = 1 kPa (abs)

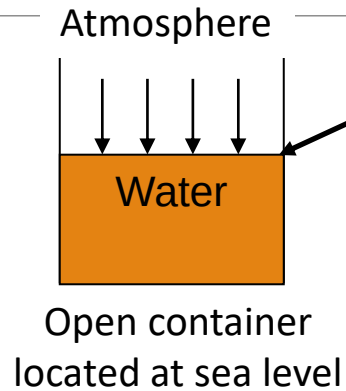
Atmospheric pressure = 101.3 kPa

Air pressure in packaging expressed in gauge pressure
 $= P_{\text{abs}} - P_{\text{atm}} = 1 - 101.3 = -100.3 \text{ kPa(g)}$

Food that is vacuum packed (located at sea level)

Example

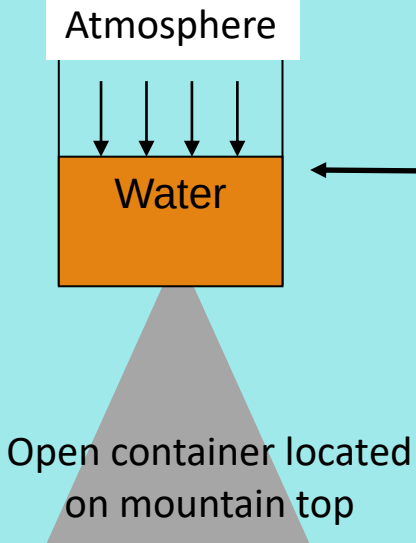
3



Pressure on water surface = atmospheric pressure
Atmospheric pressure = 101.3 kPa
Therefore pressure on water surface = 101.3 kPa (abs)

Pressure on water surface expressed in gauge pressure
$$= P_{\text{abs}} - P_{\text{atm}} = 101.3 - 101.3 = 0 \text{ kPa(g)}$$

4



Pressure on water surface = atmospheric pressure
Atmospheric pressure = 100 kPa
Therefore pressure on water surface = 100 kPa(abs)

Pressure on water surface expressed in gauge pressure
$$= P_{\text{abs}} - P_{\text{atm}} = 100 - 100 = 0 \text{ kPa(g)}$$

Absolute and Gauge Pressure



<https://sg.rs-online.com/web/>

A pressure gauge that can measure vacuum (i.e. pressure less than atmospheric pressure)

Notice the negative sign?



A pressure gauge that measure pressure higher than atmospheric pressure

Reading from pressure gauge

For example, pressure reading from this gauge is 100 psi(g)

What is this pressure in P_{absolute} given P_{atm} is 14.7 psi?

$$P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atm}}$$

Given $P_{\text{atm}} = 14.7$ psi,

$$P_{\text{absolute}} = 100 + 14.7 = 114.7 \text{ psi(abs)}$$



Expressing hydrostatic pressure in gauge pressure for fluid exposed to atmosphere

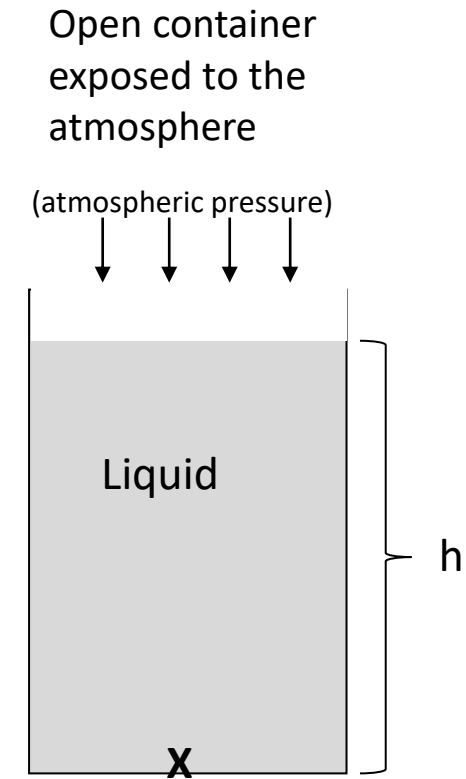
Hydrostatic pressure equation: $P = P_o + \rho gh$

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmosphere}}$$

Static pressure at Point X located at the bottom of liquid = $\rho gh + P_{\text{atm}}$

This pressure is an absolute pressure.

To convert pressure at Point X to gauge pressure, you minus P_{atm} . So the gauge pressure at Point X is simply ρgh .

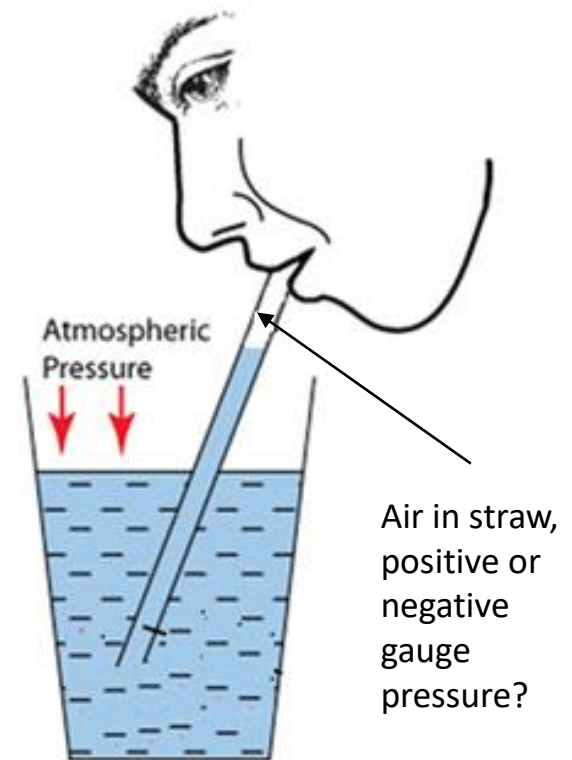


Example

The diagram shows a person hold his breath half-way while drinking.

Please indicate whether the gauge pressure is positive or negative for the air in the straw. Explain your answer using Pascal law

Hint: Which 2 points have the same pressure?



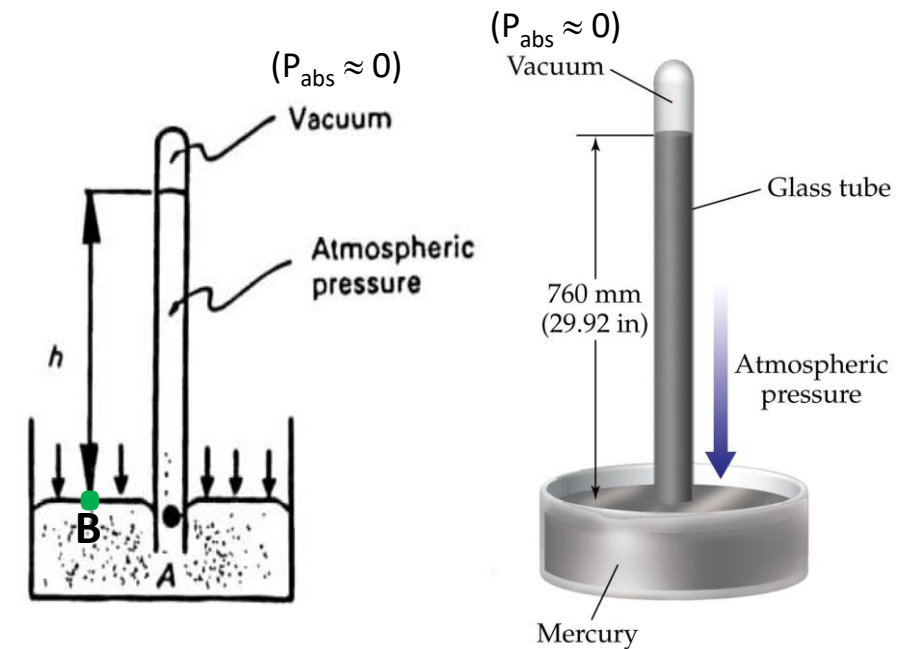
Pressure measurement

Pressure measurement

- Barometers
- Manometers
- Pressure gauges
- Pressure transducers

Barometer

- Barometer is a device that measures atmospheric pressure.
- The mercury barometer (as shown in this slide) is an old type of barometer.
 - It consists of a glass tube filled with mercury and inverted over a bath of mercury.
 - The height of the mercury column is dependent on the atmospheric pressure.
 - The higher the atmospheric pressure the higher the height of mercury column.
- Atmospheric pressure can simply be expressed as height of liquid (head of fluid as discussed earlier)



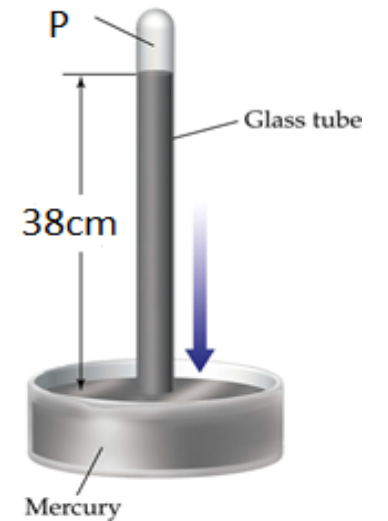
Example

A barometer reads 760 mm of mercury (S.G. of Hg is 13.54)

- A. Calculate the barometric reading (i.e. the local atmospheric pressure) in kPa(abs).
- B. If water is used instead of mercury to measure the atmospheric pressure, how high would the water column be?

Example

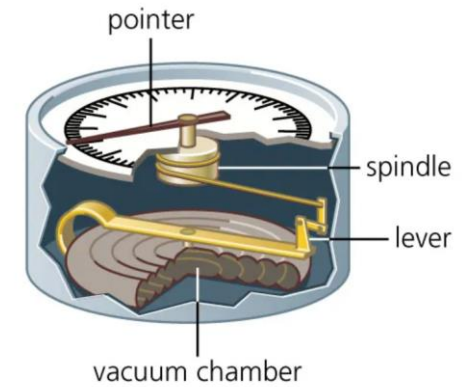
On a planet with thinner atmosphere, a mercury barometer has a reading of 38.0 cm. Assume the planet has gravitation constant, $g = 6.01 \text{ N/kg}$. (S.G. of Hg is 13.54). What is the atmospheric pressure for the planet in kPa?



Other type of barometer (for info only)

Aneroid barometer

- Does not contain liquid like mercury
- Contains an evacuated chamber which is connected to a lever and pointer, and which will move in response to changing atmospheric pressure
- Video showing how the aneroid barometer works for your information →
<https://www.youtube.com/watch?v=0xkNFN5j6Wo>

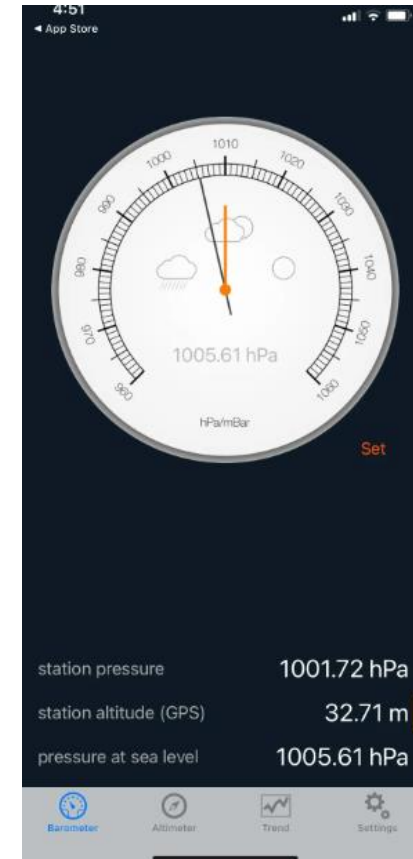


Other type of barometer (for info only)

Digital barometer

- A digital barometer contains a sensor which responds to changes in atmospheric pressure
- Many mobile phones these day have built in digital barometer.

We will not discuss how the sensor works...

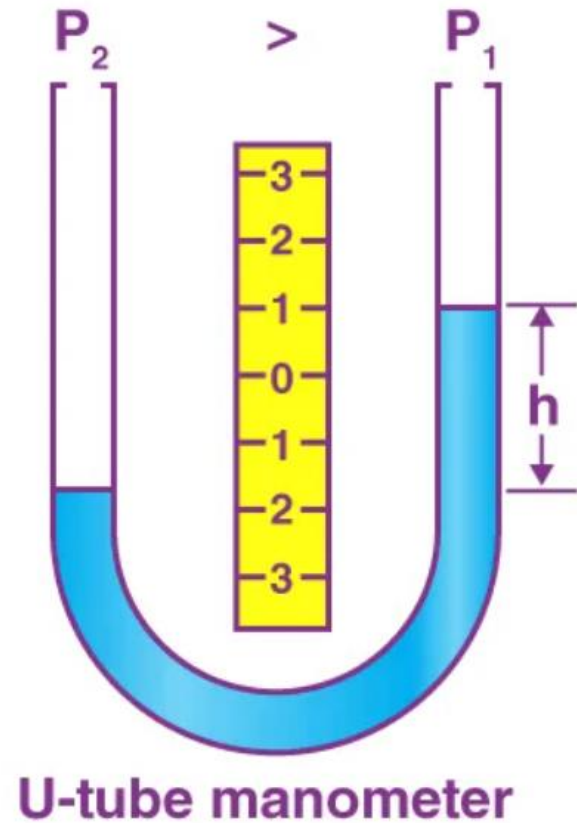


A mobile phone app which measures the atmospheric pressure

Simple Manometer

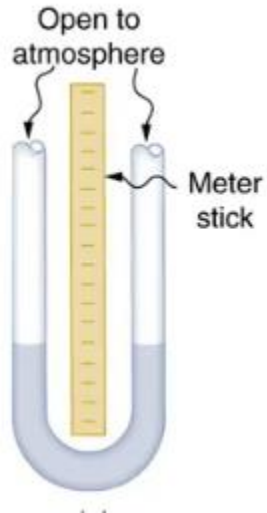
Note: in a simple manometer, the right arm of manometer is open to atmosphere, therefore $P_1 = P_{\text{atm}}$

- Manometers are pressure measuring devices which use the relationship between pressure and heights of fluid columns exerting the pressure ($P = \rho gh$)
- A simple manometer consists of a U-shaped tube having one end connected to the point where the pressure is to be measured while the other end is open to atmosphere.
- The change in fluid height can be used to determine the pressure.

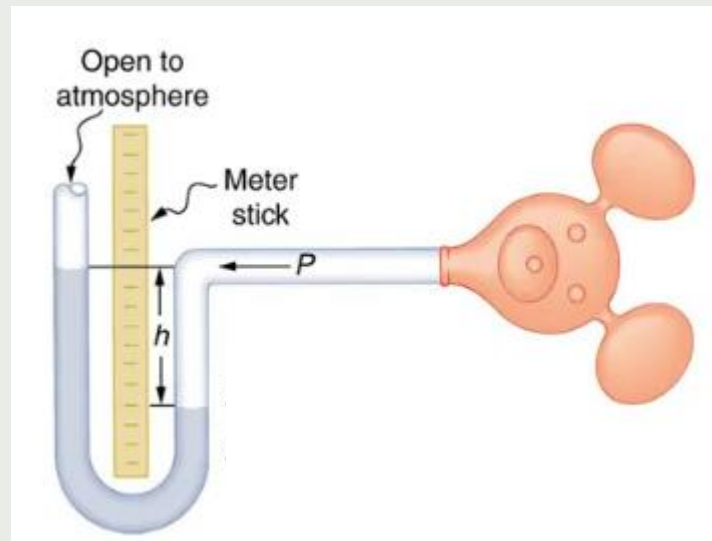


Simple Manometer

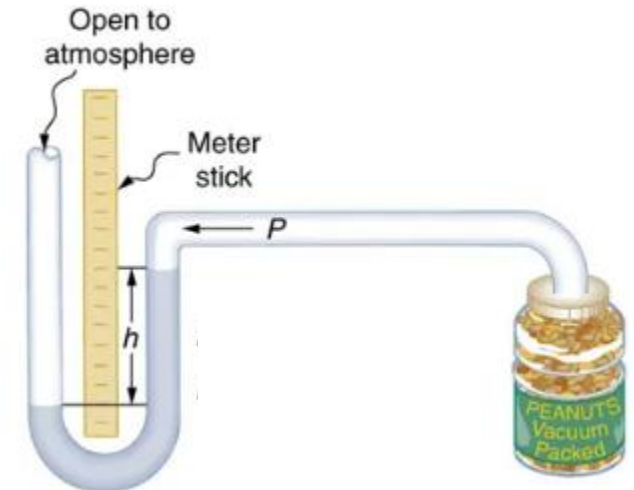
When both sides of a U-tube manometer are open to atmosphere, the atmospheric pressure pushes down on each side equally so the liquid level is the same for both sides.



When the right arm of a U-tube is connected to a point which is higher than atmosphere pressure, the fluid in the right arm is pushed down and the fluid in the left arm rises.

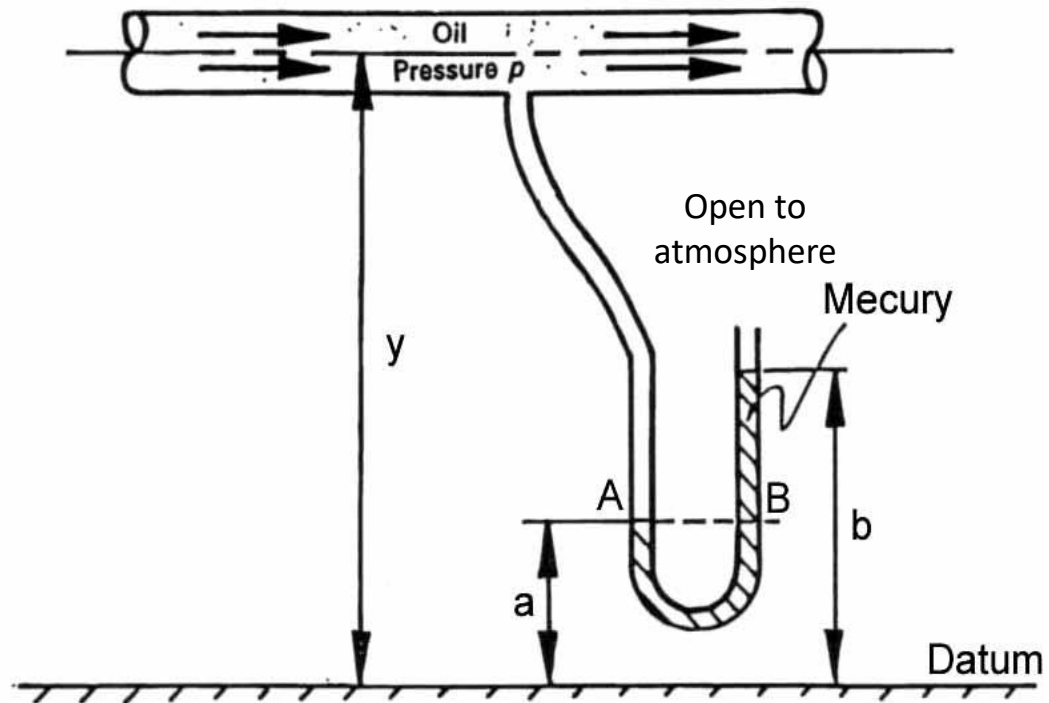


When the right arm of a U-tube is connected to vacuum, the fluid in the right arm is sucked up the tube and the fluid in the left arm drops.



Simple U-tube manometer

- Pressure can be measured by connecting one end of the U-tube manometer to a flowing fluid in a pipe or to a vessel containing fluid
- The other end of the U-tube usually open to atmosphere.



Pressure at A is :

$$P_A = P + \rho_o g (y - a)$$

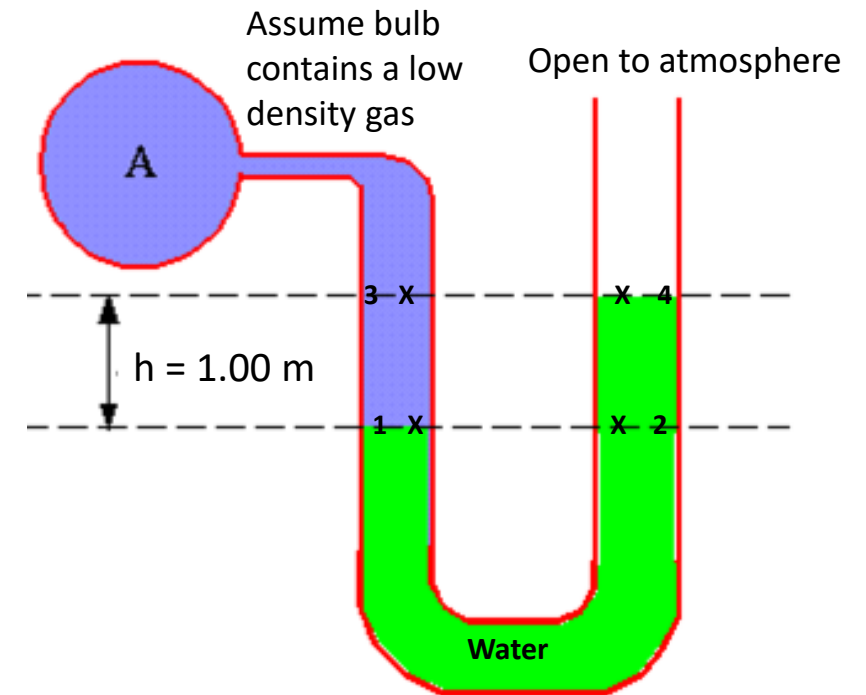
Pressure at B is :

$$P_B = \rho_{Hg} g (b - a) + P_{atm}$$

$$P_A = P_B$$

Simple manometer - how to determine pressure (Method 1)

- Suppose we want to obtain pressure of gas (P_A) by connecting a simple manometer to Point A.
- We first identify 2 points with the same pressure
(Can you correctly identify that Point 1 and 2 have the same pressure?)
- Pressure at Point 1 = Pressure of gas (P_A)
(for this question, we ignore the effect of static pressure of gas since it has a low density)
- Pressure at Point 2 = $P_{\text{atm}} + \rho_{\text{water}}gh$
- $P_A = P_{\text{atm}} + \rho_{\text{water}}gh$
- We can calculate P_A using above equation if measure h



Simple manometer - how to determine pressure

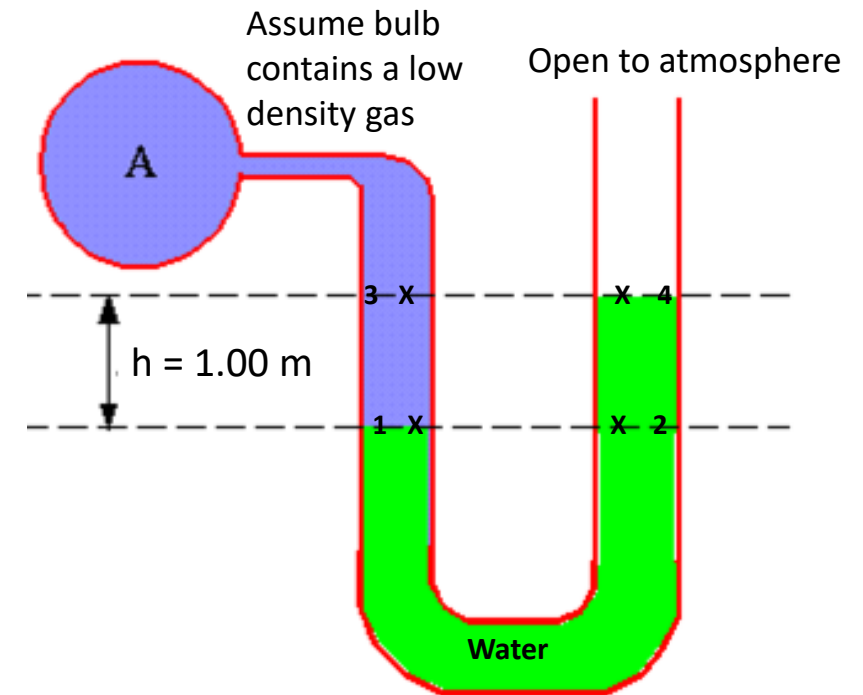
Assuming $P_{\text{atm}} = 101325 \text{ Pa}$

$P_1 = P_A$ (as explained in previous slide)

$P_2 = P_{\text{atm}} + \rho_{\text{water}}gh$

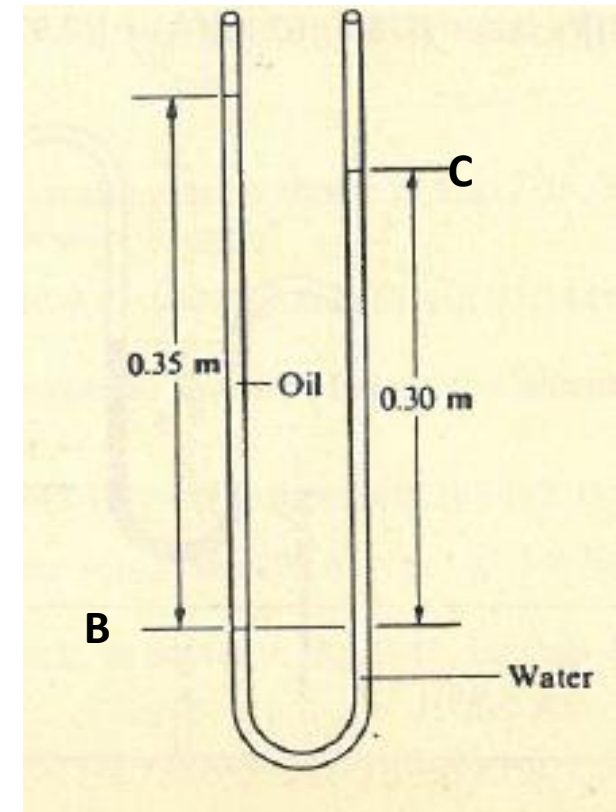
$P_1 = P_2$

→ $P_A = P_{\text{atm}} + \rho_{\text{water}}gh = 101325 \text{ Pa} + 1000 \text{ kg/m}^3 (9.81 \text{ N/kg})(1.00 \text{ m}) = 111135 \text{ Pa} = 1.11 \times 10^5 \text{ Pa}$



Example – using a U-tube to determine density of fluid

A glass U-tube open to the atmosphere at both ends is shown. If the U-tube contains oil and water, determine the specific gravity of the oil.



Solutions

Draw a horizontal line across the two points with the same pressure

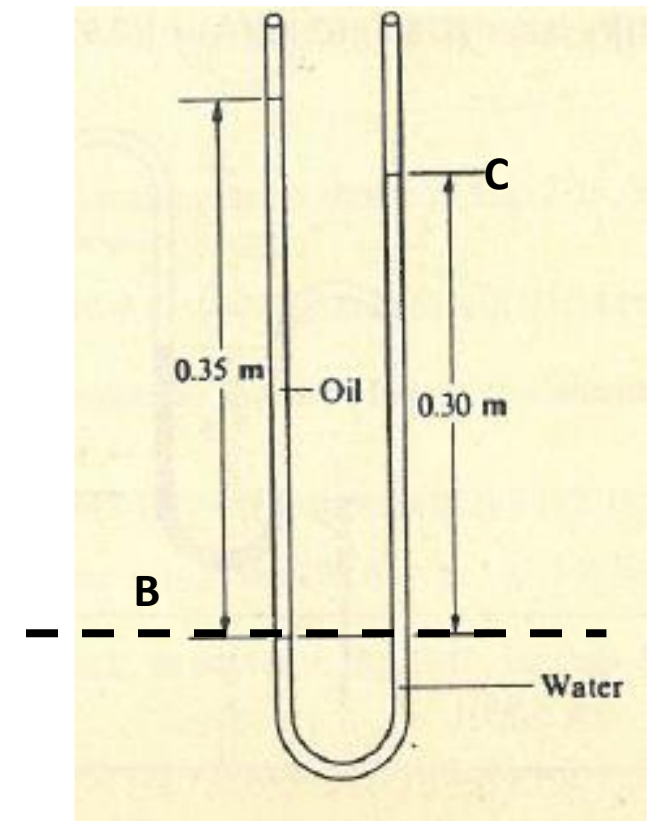
Left arm: $P_B = P_{\text{atm}} + \rho_{\text{oil}}g (0.35)$

Right arm: $P_B = P_{\text{atm}} + \rho_{\text{water}}g (0.3)$

$$P_{\text{atm}} + \rho_{\text{oil}}g (0.35) = P_{\text{atm}} + \rho_{\text{water}}g (0.3)$$

$$\rho_{\text{oil}}(9.81 \text{ N/kg})(0.35 \text{ m}) = 1000 \text{ kg/m}^3(9.81 \text{ N/kg})(0.3 \text{ m})$$

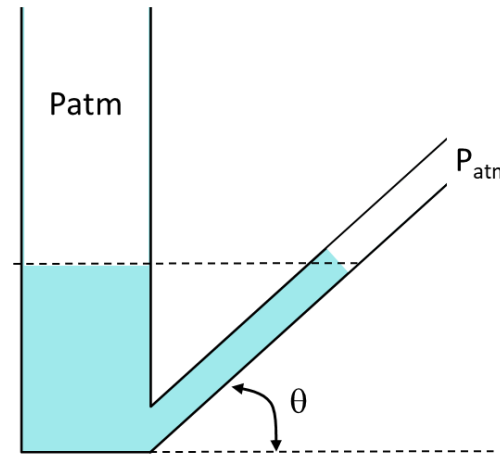
$$\rho_{\text{oil}} = 8.6 \times 10^2 \text{ kg/m}^3$$



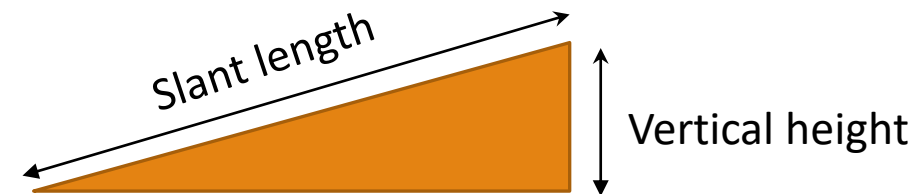
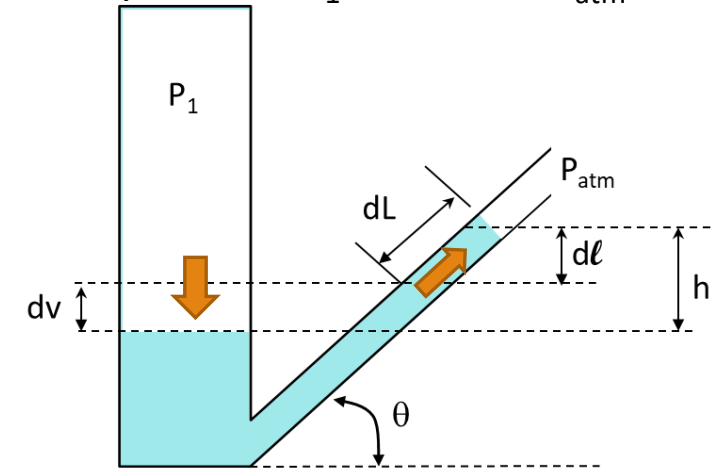
Inclined manometer

If the pressure difference to be measured is small, the leg of the U-tube may be inclined. The movement of the meniscus along the inclined leg, read off on the scale, is considerably greater than the change in height of fluid compared to when the tube is vertical.

No connected to any point to measure pressure

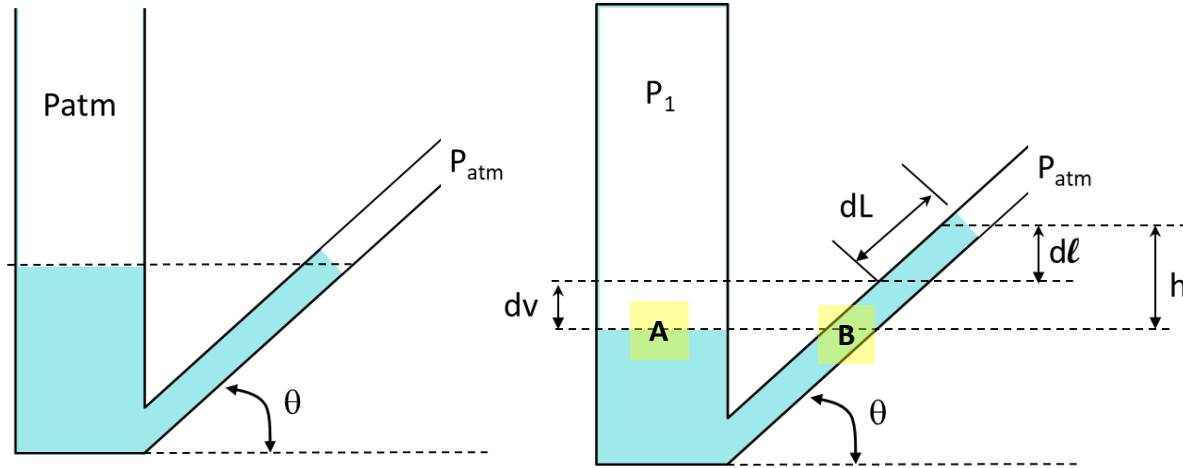


Connected to a point with pressure P_1 which is $> P_{atm}$



Slant length is always greater than vertical height. Therefore a small change in pressure will result in large movement in liquid in the inclined arm

Inclined manometer



Note:

1. Pressure at Point B is calculated based on vertical height of liquid in inclined arm and not the slanted length
2. "sin" refers to the trigonometry function sine

When inclined manometer is connected to pressure P_1 :

- The liquid in the vertical arm (left arm) will reduce in height by dv
- The liquid in the inclined arm (right arm) will increase in vertical height by $d\ell$ and slant length by dL

Therefore

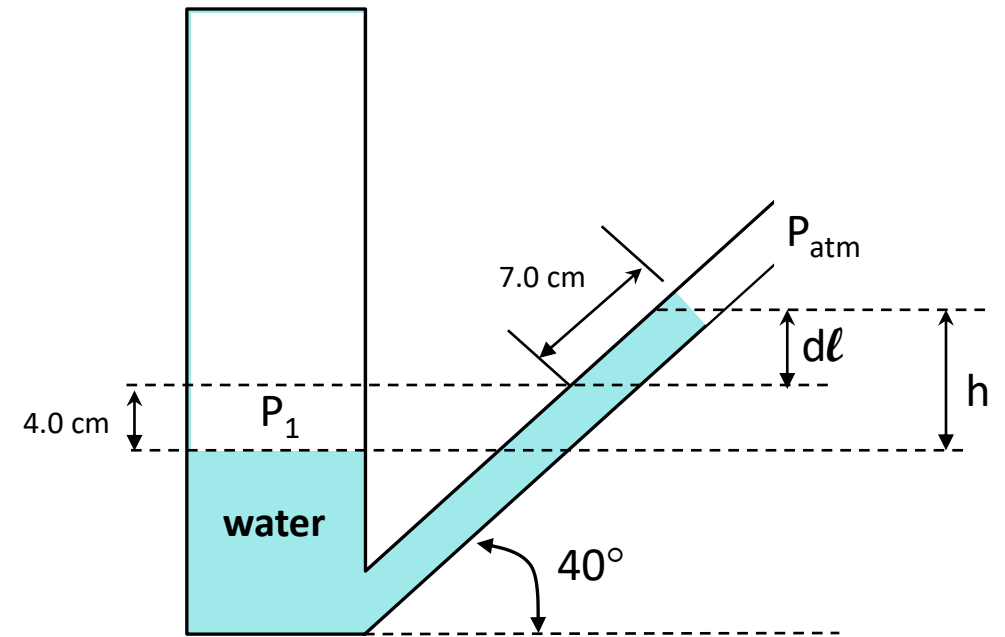
$$P_1 = P_A = P_B = P_{atm} + \rho gh = P_{atm} + \rho g(dv + d\ell)$$

where $d\ell = dL \sin(\theta)$

dL can be read off directly from the inclined manometer

Example

Find P_1 based on the readings from the inclined manometer. The liquid in the manometer is water which has a density of 1000 kg/m^3 . Express your answer in Pa (g).



Solutions

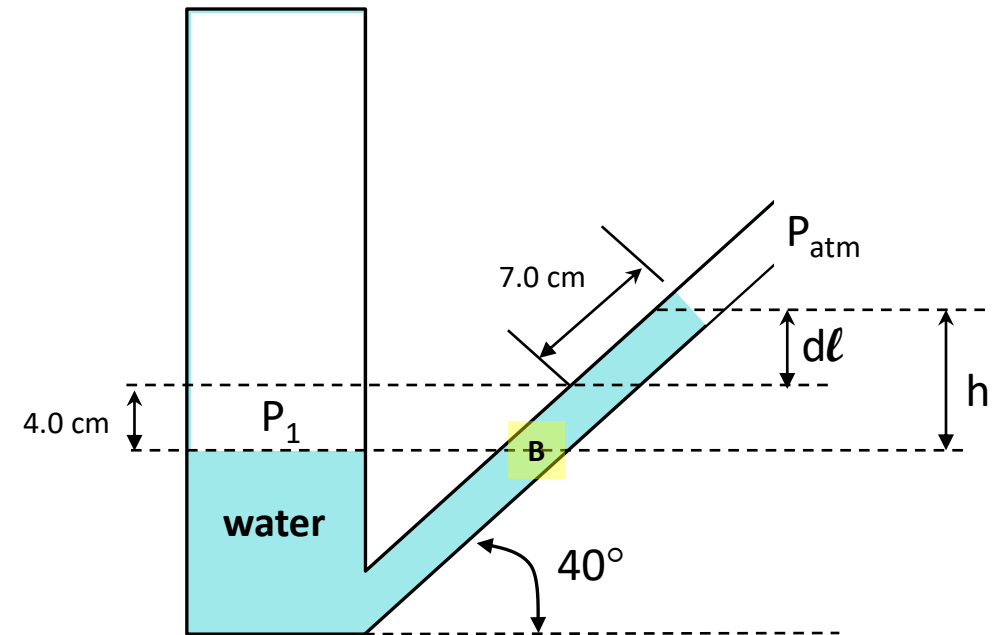
$$P_{\text{atm}} = 0 \text{ Pa (g)}$$

$$P_1 = P_B$$

$$P_B = \rho gh + P_{\text{atm}} = 1000 \text{ kg/m}^3 (9.81 \text{ N/kg})(h) + 0$$

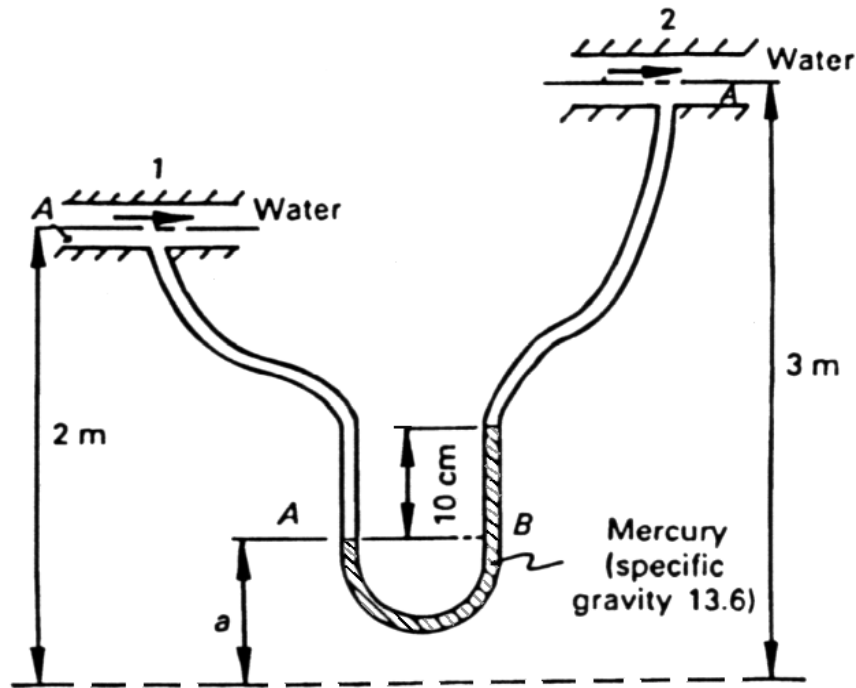
$$h = 0.040 \text{ m} + 0.070 \text{ m}(\sin 40^\circ) = 0.085 \text{ m}$$

$$P_B = 830 \text{ Pa (g)}$$



Differential manometer

- To measure the pressure different between 2 points, the manometer which is used is called the **differential manometer**.



Pressure at A is :

$$P_A = P_1 + \rho_w h_1 g \quad \text{where } h_1 = (2 - a) \text{ m}$$

Pressure at B is :

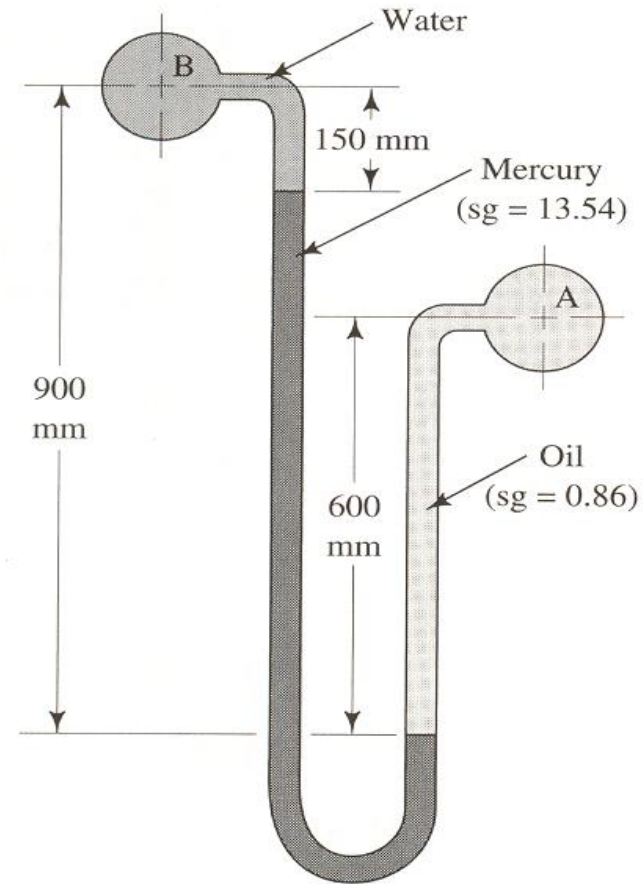
$$P_B = P_2 + \rho_w h_2 g + \rho_{Hg} h_3 g$$

$$\text{where } h_2 = (3 - 0.1 - a) \text{ m} \quad ; \quad h_3 = 0.1 \text{ m}$$

Example

For the u-tube manometer shown, calculate P_A if $P_B = 100 \text{ kPa (abs)}$.

Note this shows a differential manometer where both arms of the manometer are connected to pressure sources.



Solutions

First identify which two points have the same pressure.

$$P_E = P_D$$

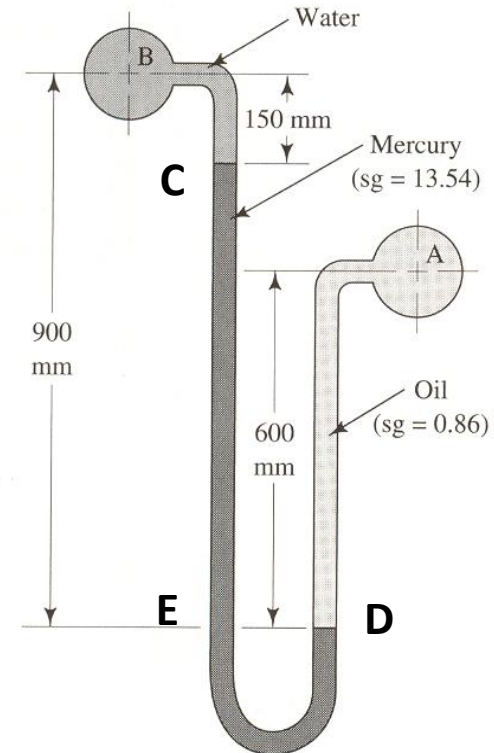
$$\text{Left side: } P_E = P_B + \rho_{\text{water}}g(0.15\text{m}) + \rho_{\text{Hg}}g(0.9 - 0.15)\text{m}$$

$$\text{Right side: } P_D = P_A + \rho_{\text{oil}}g(0.6\text{m})$$

$$\text{Therefore } 100 \times 10^3 + \rho_{\text{water}}g(0.15\text{m}) + \rho_{\text{Hg}}g(0.9 - 0.15) = P_A + \rho_{\text{oil}}g(0.6\text{m})$$

Solve for P_A

$$P_A = 196 \text{ kPa (abs)}$$

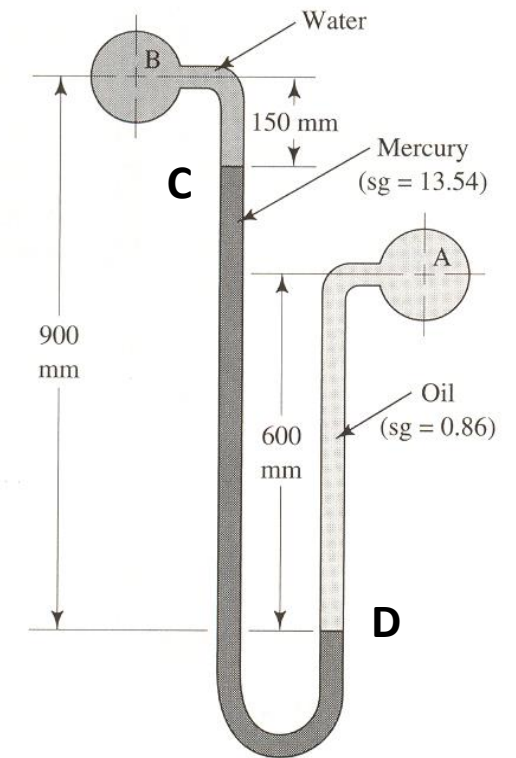


Solving manometer problem using another method (Method 2)

For the u-tube manometer shown, calculate P_A if $P_B = 100\text{kPa(abs)}$. We will solve this same question using another method that involves calculating the pressure difference between 2 adjacent interfaces.

Description of method:

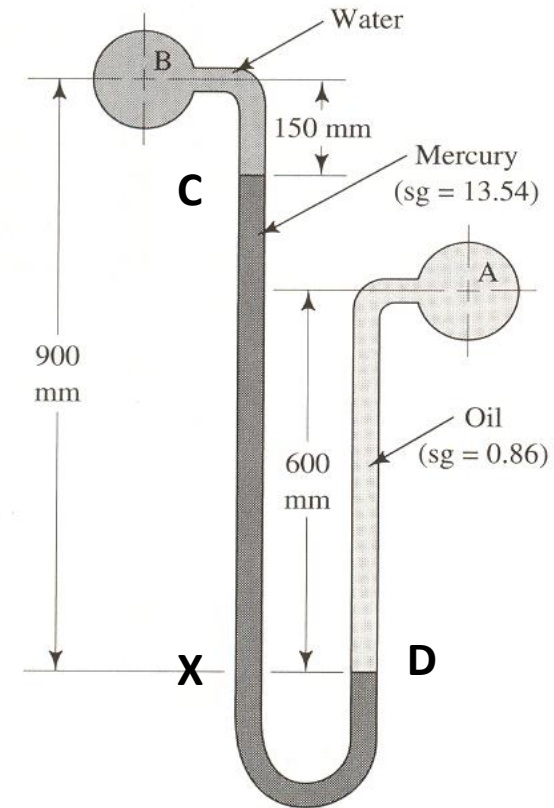
1. Locate all the interface
2. Start from left side of the U-tube and proceed to calculate the pressure difference between two adjacent interfaces. i.e. $P_C - P_B$, $P_D - P_C$, $P_A - P_D$
(You can also do in reverse direction, try it out, but must be consistent)
3. Add up all the pressure difference and you will find all intermediate terms cancel away to give you the answer you need...



Solving manometer problem using another method

How to determine the pressure difference between 2 adjacent interfaces

1. Use ρgh to help you find the pressure difference between 2 points e.g. $P_C - P_B = \rho_{\text{water}} g (0.15\text{m})$
2. When second point is higher than first point, e.g. $P_A - P_D$, what must you be careful about?



Solving manometer problem using another method

Which one higher pressure?		Answer
P_B	P_C	P_C
P_C	P_D	P_D
P_D	P_A	P_D

$$P_C = P_B + \rho_{\text{water}} g (0.15\text{m})$$

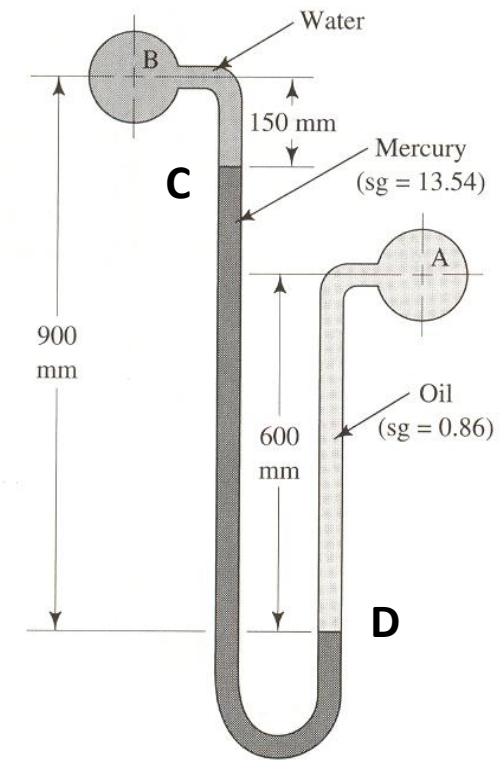
$$P_C - P_B = \rho_{\text{water}} g (0.15\text{m}) \text{ ---- (1)}$$

$$P_D = P_C + \rho_{\text{Hg}} g (0.9\text{m} - 0.15\text{m})$$

$$P_D - P_C = \rho_{\text{Hg}} g (0.9\text{m} - 0.15\text{m}) \text{ ---- (2)}$$

$$P_D = P_A + \rho_{\text{oil}} g (0.6\text{m})$$

$$P_A - P_D = -\rho_{\text{oil}} g (0.6\text{m}) \text{ ---- (3)}$$



Solving manometer problem using another method

$$P_C - P_B = \rho_{\text{water}} g (0.15\text{m}) \text{ ---- (1)}$$

$$P_D - P_C = \rho_{\text{Hg}} g (0.9\text{m} - 0.15\text{m}) \text{ ---- (2)}$$

$$P_A - P_D = -\rho_{\text{oil}} g (0.6\text{m}) \text{ ---- (3)}$$

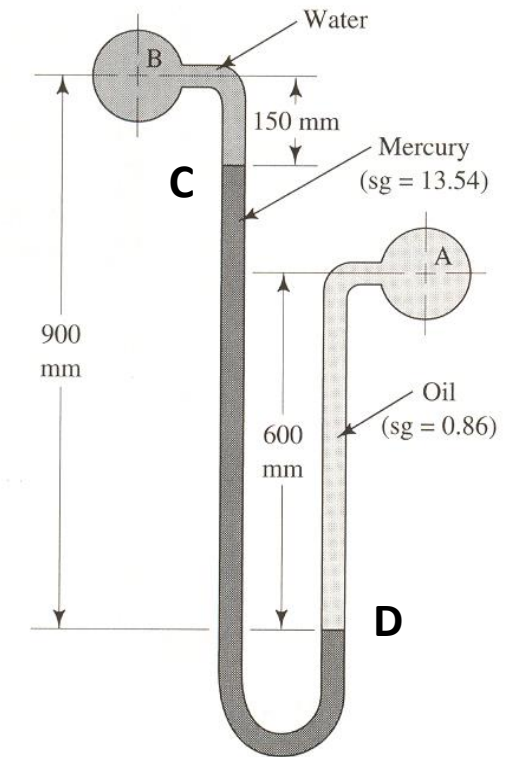
What happen when you add up equation 1, 2 & 3?

Only $P_A - P_B$ is left

Therefore

$$P_A - P_B = \rho_{\text{water}} g (0.15\text{m}) + \rho_{\text{Hg}} g (0.9 - 0.15) - \rho_{\text{oil}} g (0.6\text{m})$$

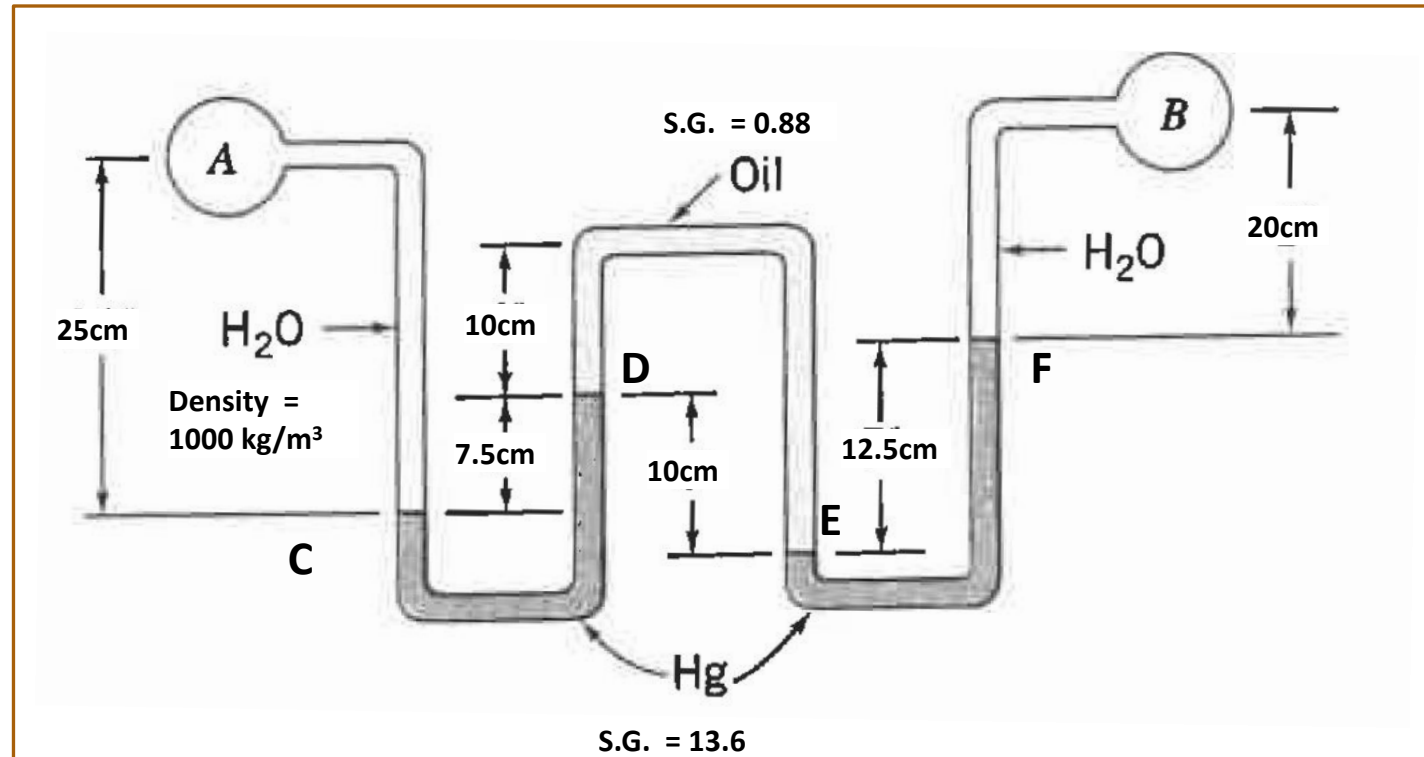
You can therefore calculate P_A since P_B is 100kPa (abs)



Example – multi-tube manometer

Can you try to use Method 2 to determine $P_A - P_B$?

You can also use Method 1, but it would be more tedious. Method 2 is preferred when you encounter problems involving multi-tube manometer.



Solutions - partial

We are going to calculate the pressure difference from A to B

You can either do:

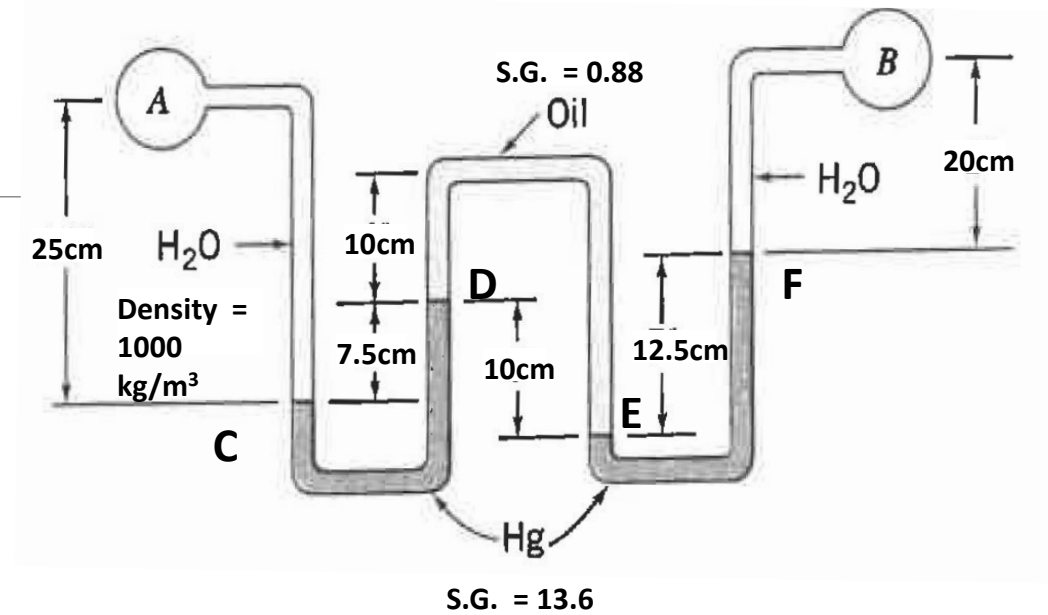
- $P_C - P_A$
- $P_D - P_C$
- $P_E - P_D$
- $P_F - P_E$
- $P_B - P_F$

The sum of all these terms is $P_B - P_A$

Or

- $P_A - P_C$
- $P_C - P_D$
- $P_D - P_E$
- $P_E - P_F$
- $P_F - P_B$

The sum of all these terms is $P_A - P_B$



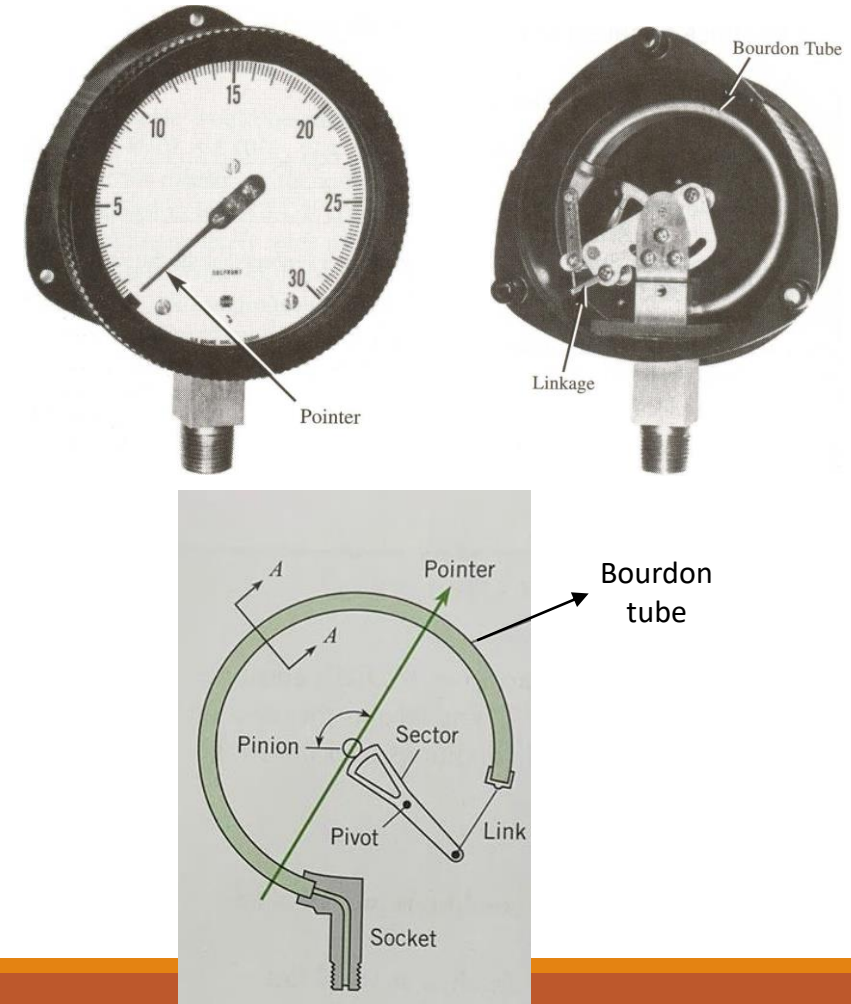
Do you get the answer $P_A - P_B = 25.3 \text{ kPa}$?

Bourdon tube pressure gauge

Bourdon tube pressure gauge

- Measures pressure by sensing the deflection of a coiled tube.
- When pressure is applied to the gauge, the curved tube tends to straighten, causing the pointer to move to give a pressure reading.
- The tube is undeflected when it is experiencing atmospheric pressure. At this condition, the gauge pointer is calibrated to read zero pressure.
(think of gauge pressure = 0).

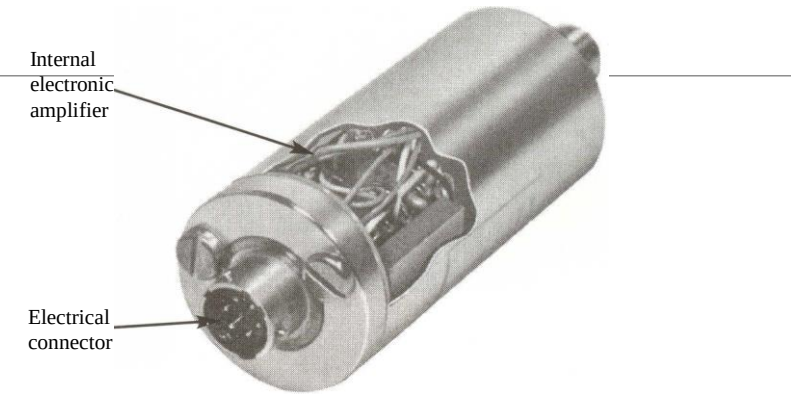
<https://www.youtube.com/watch?v=84Vxv1GBGmM>



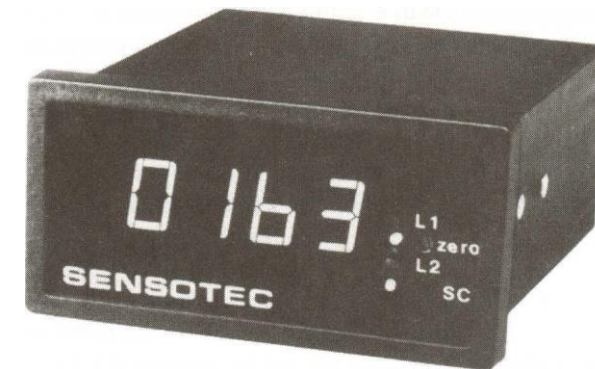
Pressure transducers

- Measures some physical quantity and generates an electrical signal that has a predictable relationship to the measured quantity.
- The level of the electrical signal then can be recorded, displayed on a meter, or stored in a computer memory for later display or analysis.

<https://www.youtube.com/watch?v=UZLiLRljzbU>



Strain gage pressure transducer

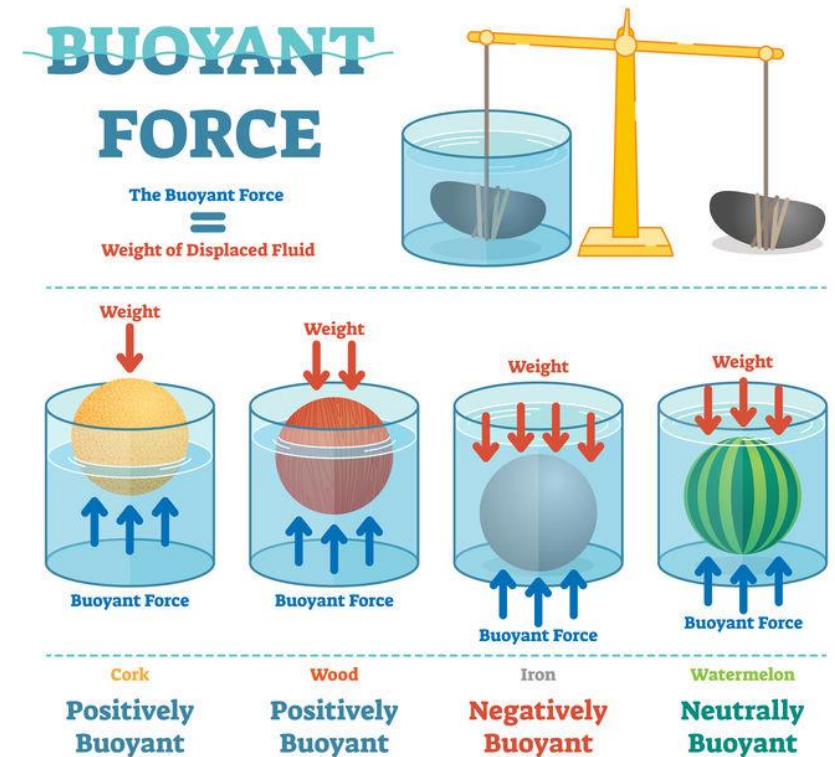


Digital electronic amplifier/indicator

Archimedes' Principle

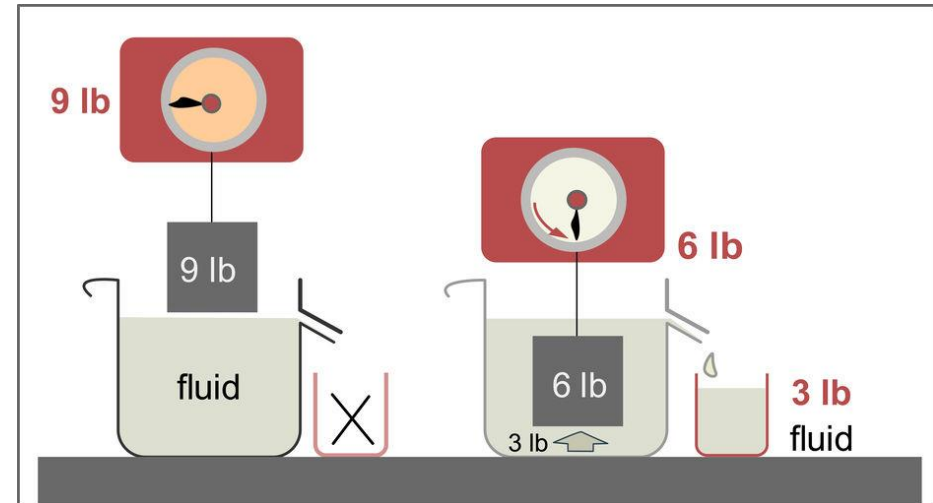
Buoyancy

- Buoyancy (a.k.a. as buoyant force or upthrust) , B or F_B is the force that is exerted on an object that is either wholly or partially immersed in a fluid.
- It acts upward for most of the daily situations we encountered e.g. helium filled balloon floats in air
- It is caused by differences in pressure acting on opposite sides of an object immersed in a static fluid.
- SI unit is Newton (N)



Archimedes' Principle

Any object completely or partially immersed in a fluid experiences an upward force (also known as buoyant force) which is equal to the weight of the fluid displaced by the object.



The example shows an object with a mass of 9 lb weighing only 6 lb when placed in water because there is a buoyant force which is equivalent to the weight of 3 lb of fluid acting on the object.

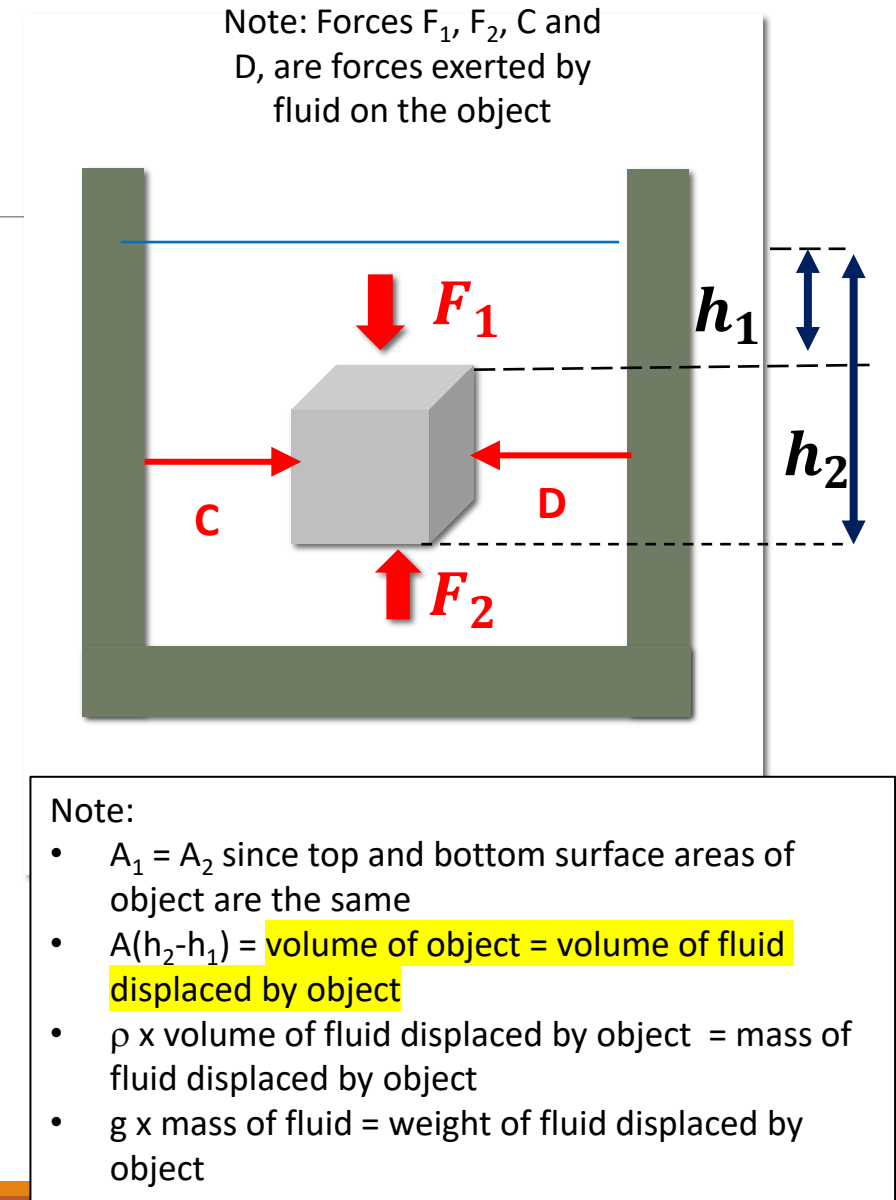
Deriving Archimedes' Principle

Consider the forces acting on an object by the fluid which has density of ρ :

- Forces C and D cancel out each other $\rightarrow$ the object is not moving left or right
- F_1 is the force acting on the top surface of the object by the fluid = $P_1 A_1$ (where A_1 = top surface area of object)
- F_2 is the force acting on the bottom surface of the object by the fluid = $P_2 A_2$ (where A_2 = bottom surface area of object)
- $F_2 > F_1$ since $P_2 > P_1$ (because $P_1 = \rho g h_1$ & $P_2 = \rho g h_2$; $h_2 > h_1$)

The difference in pressure between the top and bottom surface results in a net upward force. This is the buoyant force or upthrust.

- **Buoyant force (F_B)** = $F_2 - F_1 = P_2 A_2 - P_1 A_1 = \rho g h_2 A_2 - \rho g h_1 A_1 = \rho g A (h_2 - h_1)$
= $\rho g \times \text{Volume of object}$ = **weight of fluid displaced by object** = W_f

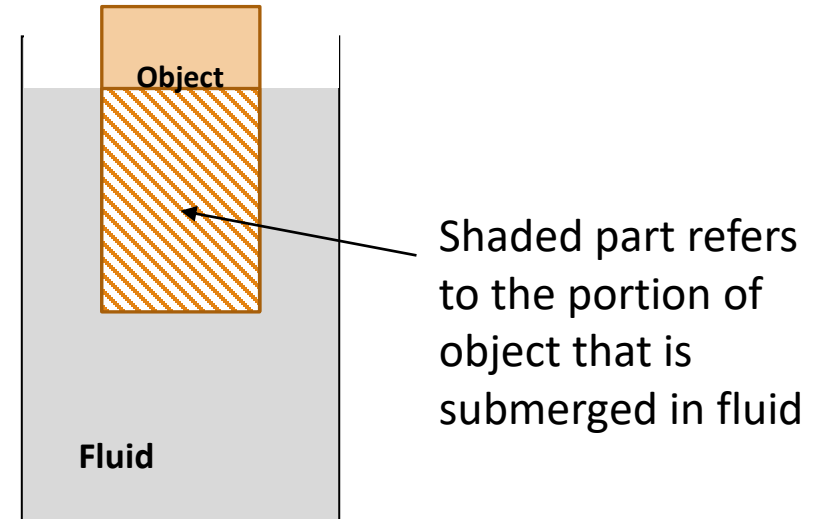


What is the meaning of weight of fluid displaced by object?

Volume of fluid displaced by object = Volume of object that is submerged in fluid

For example, if volume of the submerged portion of the object is 100 cm^3

Weight of fluid displaced by object = Weight of 100 cm^3 of fluid



Does the object rise or sink?

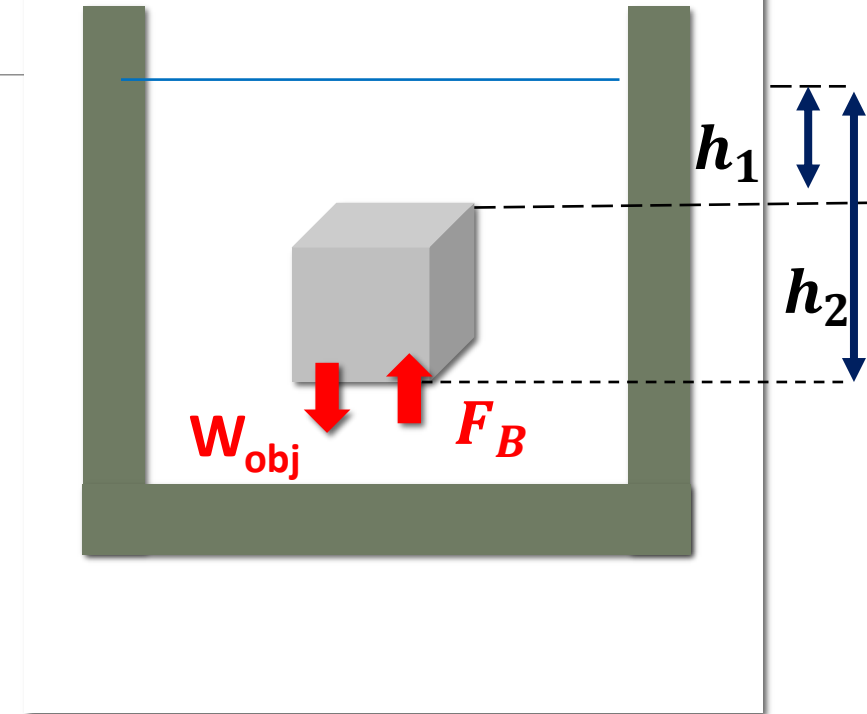
Archimedes' Principle states that the buoyancy force = weight of fluid displaced by object

$$F_B = W_f$$

What happen when we compares F_B to weight of object (W_{obj})

- When $F_B = W_f < W_{obj}$, object sinks in the fluid until it touches the bottom of the container
- When $F_B = W_f > W_{obj}$, object rises in the fluid until the weight of the fluid displaced by object equal to W_{obj} . As the object starts rising in the fluid, a greater portion of the object will emerge outside the fluid, hence the weight of fluid displaced will reduce until it is equal to W_{obj} . Then the object will stop rising. Imagine pushing a cube of ice into water and observe the ice cube rising up the water until it stops.

Note: Force F_B is the buoyant force and Force W is the weight of object

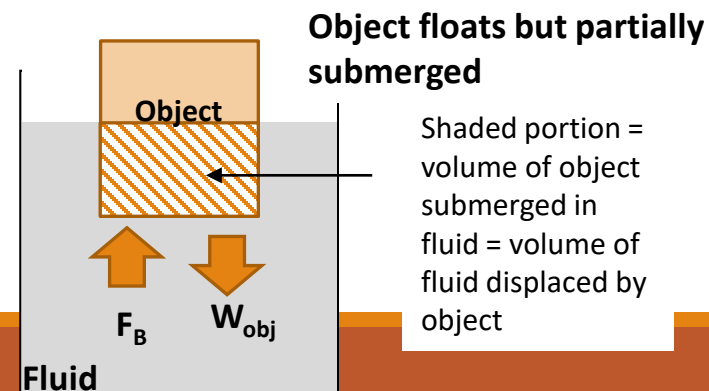
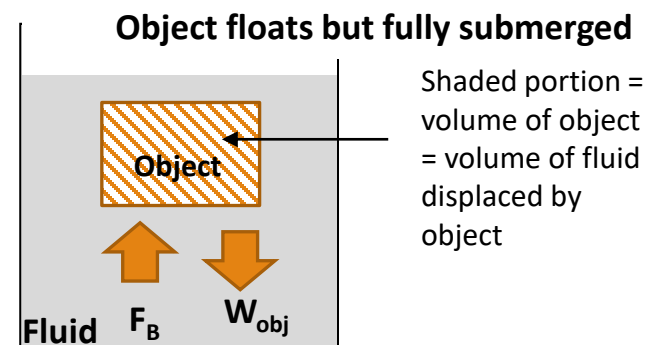
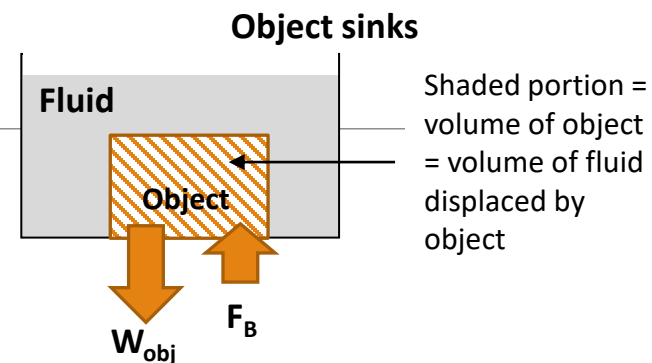


Note: When an object floats in water,
 $F_B = W_f = W_{obj}$

Application of Archimedes' principle (In Summary)

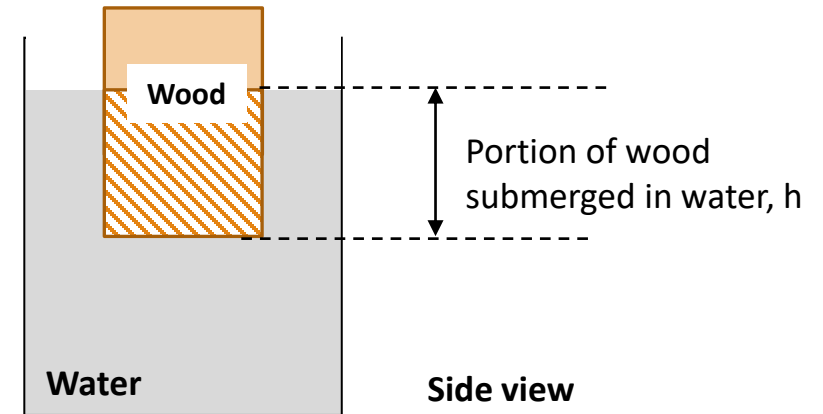
Scenarios	Relationship between F_B , W_f and W_{obj}	Relationship between V_{obj} and $V_{fluid\ displaced}$
Object sinks	$F_B = W_f < W_{obj}$	$V_{obj} = V_{fluid\ displaced}$
Object floats in fluid but fully submerged	$F_B = W_f = W_{obj}$	$V_{obj} = V_{fluid\ displaced}$
Object floats in fluid but partially submerged	$F_B = W_f = W_{obj}$	$V_{obj} > V_{fluid\ displaced}$ $V_{obj\ submerged\ in\ fluid} = V_{fluid\ displaced}$

Where: F_B = buoyant force, W_f = weight of fluid displaced by object, W_{obj} = weight of object, V_{obj} = volume of object, $V_{obj\ submerged\ in\ fluid}$ = volume of object submerged in fluid, $V_{fluid\ displaced}$ = volume of fluid displaced by object



Example

A rectangular block of wood has a density of $6.00 \times 10^2 \text{ kg/m}^3$. Its top surface area is 5.70 m^2 and its volume is 0.60 m^3 . The wood is placed in water. What is the depth of the wood that is submerged in the water? Water density = $1.00 \times 10^3 \text{ kg/m}^3$.



Solution

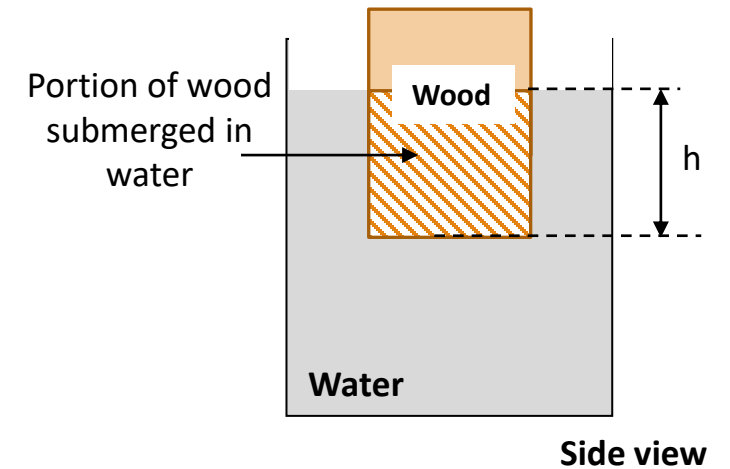
Recall that in this case where object float but partially submerged, $F_B = W_f = W_{\text{obj}}$ and $V_{\text{fluid displaced}} = V_{\text{obj submerged in fluid}}$

$W_f = \text{Weight of water displaced by wood} = 5.7h \times 1000 \text{ kg/m}^3 \times 9.81 \text{ N/kg} = (55917h) \text{ N}$ [Recall $m = \rho V$ and $W = mg \rightarrow W = \rho Vg$]

$W_{\text{obj}} = \text{Weight of wood} = m_{\text{wood}} \times g = \rho_{\text{wood}} V_{\text{wood}} g = 6.00 \times 10^2 \text{ kg/m}^3 \times 0.60 \text{ m}^3 \times 9.81 \text{ N/kg} = 3531.6 \text{ N}$

According to Archimedes' principles, $F_B = W_f$ therefore:

$3531.6 = 55917h \rightarrow h = 0.063 \text{ m}$



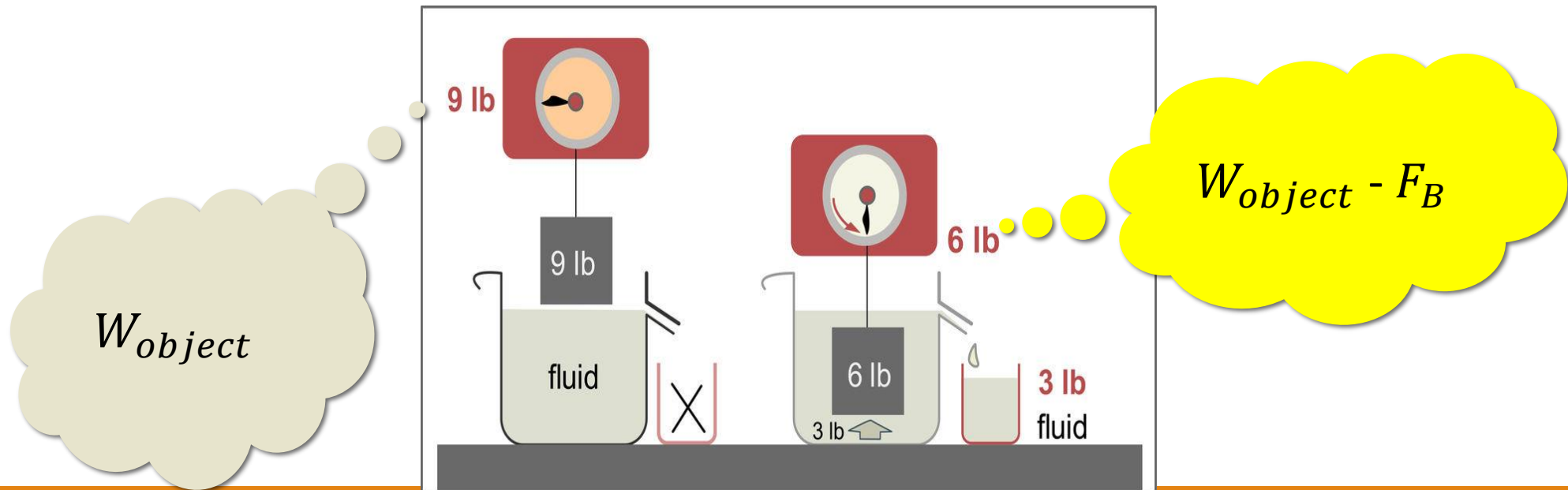
Apparent weight

The weight of an object is affected by the buoyancy of fluid in which it is immersed.

Apparent weight = net downward force

$$= W_{object} - F_B$$

So, $F_B = \text{Weight} - \text{apparent weight} = \text{weight of liquid displaced by object}$



Example

A gold statue has a mass of 109 kg in air and an apparent mass of 58kg when completely submerged in water.

Given:

Density of water = $1,000 \text{ kg/m}^3$, $g = 9.81 \text{ N/kg}$

- a) Calculate the volume of the statue.
- b) Calculate the density of the statue.

Solutions

$$F_B = \text{Weight} - \text{apparent weight} = \text{weight of liquid displaced by object } (W_f)$$

A. Calculate the volume of the statue.

$$F_B = \text{weight} - \text{apparent weight} = (109 - 58) \text{ kg} \times 9.81 \text{ N/kg} = 500.31 \text{ N}$$

So, weight of water displaced (W_f) is 500.31 N

Since the statue is completely submerged in the water, the volume of water displaced = volume of statue

$$\text{Since } \rho = \frac{m}{V} \rightarrow m = \rho V \rightarrow \text{Weight, } W_f = m_{\text{water}} g = \rho_{\text{water}} V_{\text{water displaced}} g$$

$$500.31 \text{ N} = 1000 \text{ kg/m}^3 \times 9.81 \text{ N/kg} \times V$$

$$V = 0.051 \text{ m}^3 \text{ [This is the volume of water displaced, and it is also equal to the volume of the statue]}$$

B. Calculate the density of the statue.

$$\text{Density of statue} = \text{mass of statue} / \text{volume of statue} \rightarrow 109 \text{ kg} / 0.051 \text{ m}^3 = 2100 \text{ kg/m}^3$$